

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

July 2022

Course Code: EIE101R01

Course: **BASIC ELECTRONICS ENGINEERING**

Question Paper No. :U0112

Duration: 3 hours

Max. Marks:100

PART- A

Answer all the questions

10 x 2 = 20 Marks

1. An NPN bipolar junction transistor (BJT) is operating in the active region. If the reverse bias across the base-collector junction is increased, what happens to the width of the base and the emitter current?
2. Draw the load line for the given circuit shown in Figure 1.

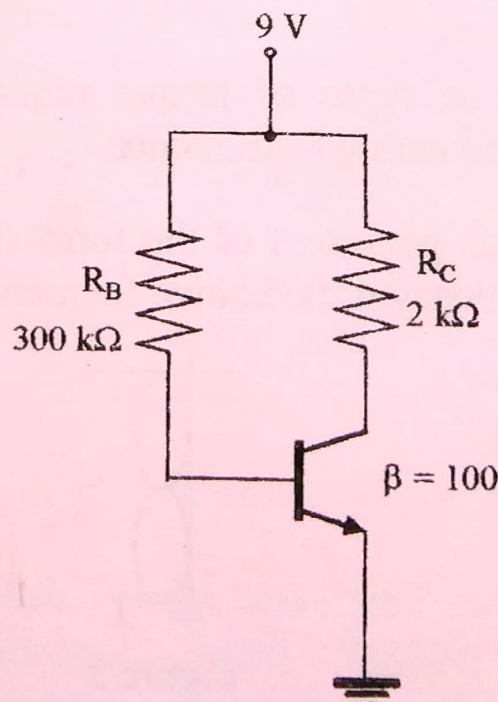


Figure 1

3. Where should the Operating point be located in the load line? Why? What happens if the location is not chosen properly?
4. Relate the leakage current in the common base and common emitter transistor circuit. Sketch the common collector transistor configuration.
5. Determine the load current I_L and Zener diode current I_Z for the circuit shown in Figure 2. The breakdown voltage of the Zener diode is 5V. Given $V_S = 20\text{ V}$, $R_S = 1\text{ k}\Omega$ and $R_L = 1.2\text{ k}\Omega$.

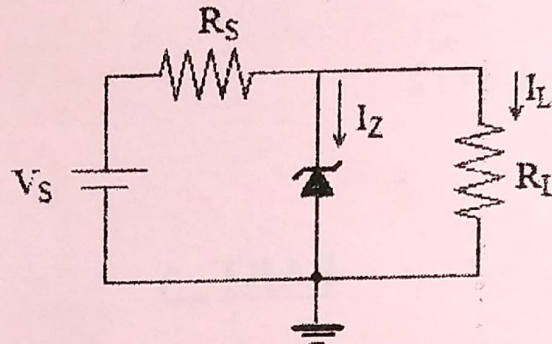


Figure 2

6. List two differences between Light Emitting Diode and Photodiode.
7. Evaluate the amplification factor if the drain resistance is $20\text{ k}\Omega$ and trans-conductance is 0.4 mS .
8. Solve for the drain current if $V_{GS} = 3\text{ V}$, $k_n = 0.3\text{ mA/V}$ and $V_{TN} = 1\text{ V}$.
9. Name the three types of torque required to ensure the proper operation of indicating instruments.
10. If the Lissajous pattern is of the form shown in Figure 3, find the unknown frequency if the known frequency is 3 kHz .

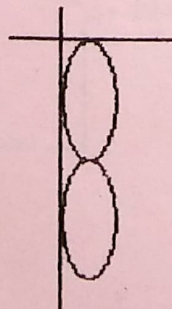


Figure 3

PART- B

Answer any four questions

4 x 15 = 60 Marks

11. (a) (i) It is necessary to clip all signals above +2 V using a combination of the diode, resistor and a battery. Outline the circuit diagram and explain the clipping action along with the input and output waveforms. (ii) If all signal below +2 V is to be clipped how will the circuit get modified? Outline the circuit diagram and explain the clipping operation along with the input and output waveforms. (8)
- (b) In a Full-wave rectifier the voltage applied to each diode is $240 \sin (377t)$. The load resistance is $R_L = 2000 \Omega$ and each diode has a forward resistance of 400Ω . Determine
- (i) Peak value of current
 - (ii) dc value of current
 - (iii) RMS value of current
 - (iv) Rectifier efficiency
 - (v) Ripple factor
 - (vi) PIV
 - (vii) Output ripple frequency (7)
12. (a) Develop the relationship between alpha and beta. (5)
- (b) Plan the typical set-up of the transistor configuration which has high current gain and high voltage gain and explain the input-output characteristics. (10)
13. (a) Explain with a neat diagram the voltage-controlled device which operates for $V_{GS} = 0V$, $V_{GS} = -1V$ and $V_{GS} = +2 V$. Plot the transfer characteristics and explain how you get the curve. (8)
- (b) Explain how Complementary Metal Oxide Semiconductor (CMOS) is used as an inverter. (7)
14. (a) Discuss the working of Permanent Magnet Moving Coil with a neat diagram. (7)
- (b) Determine the output voltage and Zener current in a transistor series voltage regulator circuit. Assume $R_L = 1 k\Omega$. The

unregulated input is 20 V, the resistor connecting the base and collector $R = 220\ \Omega$, $V_Z = 12\text{ V}$ and $\beta = 50$. (8)

15. With a neat sketch explain how the signal from ac mains with a frequency of 60 Hz can be converted to DC using four diodes. Also, develop the expression for Average current, RMS current, ripple factor, input power, output power and efficiency.
16. (a) Explain with a neat diagram the Basic DC ammeter circuit and derive the expression for shunt resistor R_{sh} . (10)
- (b) A moving coil meter has a fixed shunt of $0.01\ \Omega$ with a coil resistance of $600\ \Omega$ and a voltage drop of 350 mV across it, the full-scale deflection is obtained. Evaluate
- (i) I_{shunt} (Shunt current)
- (ii) R_m to give a full-scale deflection if $I_{sh} = 40\text{ A}$. (5)

PART- C

Answer the following

1 x 20 = 20 Marks

17. (a) Choose the proper turn ratio of the transformer if the primary voltage is 230 V_{rms} and the peak secondary voltage is 32.53 V in a regulated power supply. (2)
- (b) Explain the working of the NPN transistor used as a regulator in regulated power supply. (5)
- (c) Why is FET widely preferred over BJT? List the parameters of FET. (5)
- (d) Explain with a neat block diagram the regulated power supply and describe how regulation is achieved when there is fluctuation in the input line voltage. (8)
